SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Concept Mapping

Course Code:	CSA0311	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis		
Student Name:	ROHITH S	Reg. No.:	192421276
Section / Batch:	B.TECH IT	Date:	13/07/2025

Code of Conduct

I ROHITH S (192421276) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Rohith

Instructions

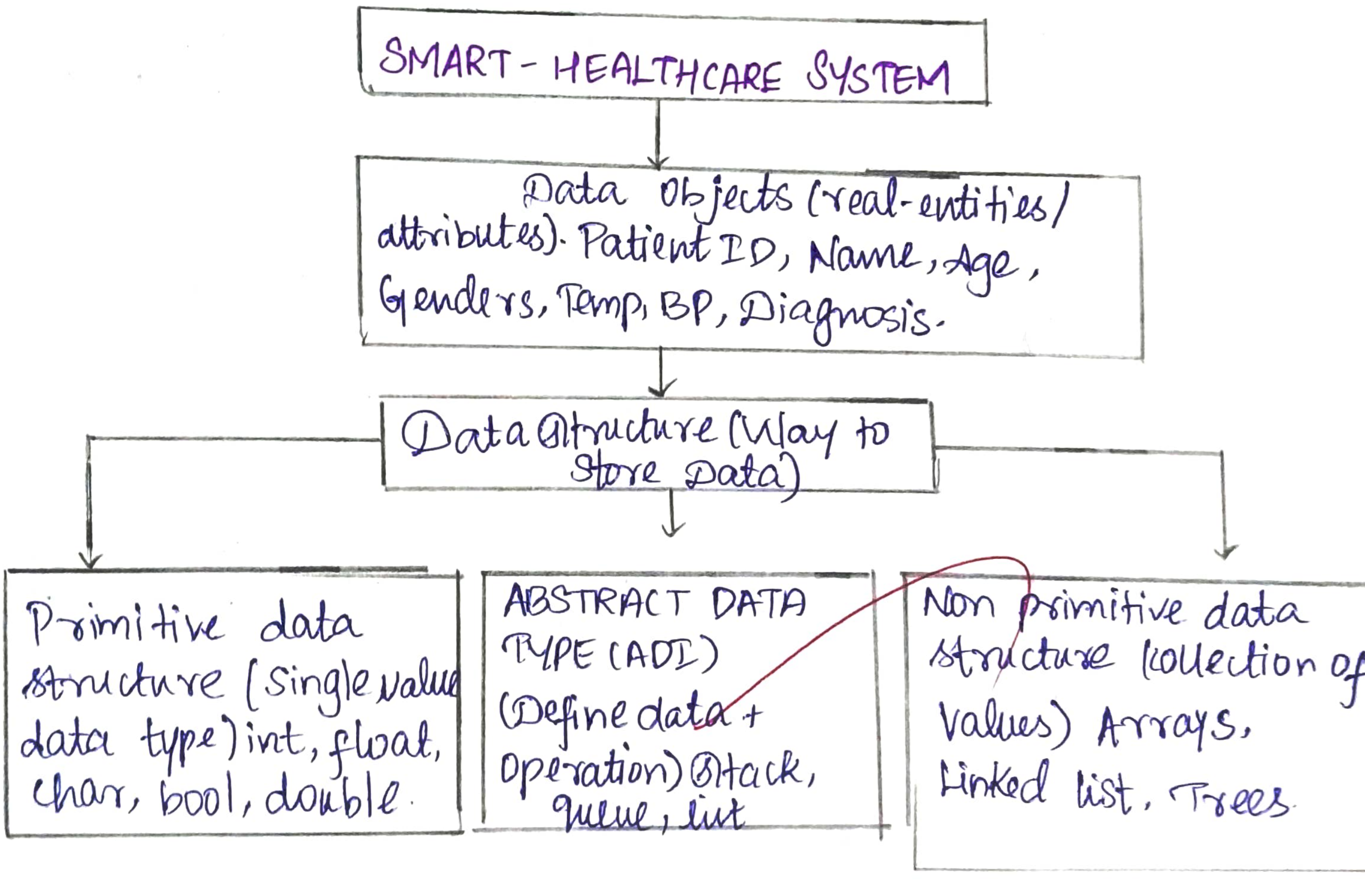
- This test contains 2 concept mapping. Total: 100 Marks.
- Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification , Linkages, Organization, Integration of Literature, and Clarity
- No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

Rubrics

Questions	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Total Marks
Q1						
Q2						

Arun
C812

Q1] A Smart health care system processes large volumes of patient data such as temperature readings, medical history and diagnostic details. Construct a concept map linking the progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant sub-concepts such as structures, pointers and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how level contributes to efficient data organization, processing and improved system performance.



STACK
(LIFO - Last in First out)
→ used for undo operation

OPERATIONS
Push()
Pop()
Peek()
isEmpty()

OPERATION
Enqueue()
Dequeue()
Front()
isEmpty()

QUEUE
(FIFO - First in First out)
→ used for patient flow management

OPERATION
Insert()
Delete()
Search()
Traverse()

LIST
(Linear collect) → used for storing patient records

↓ Implemented Using

Implementation In C - Language			
Structure	Pointers	Functions	Dynamic Memory

struct Patient{
int id;
char name[50];
int age;
float temp;
. . . .
}
	used to link nodes in linked list /Tree
	struct node next;
		• Insert()
		• Delete()
		• Search()
		• Update()
		• Display()
			• malloc()
			• calloc()
			• free()

↓ Leads to

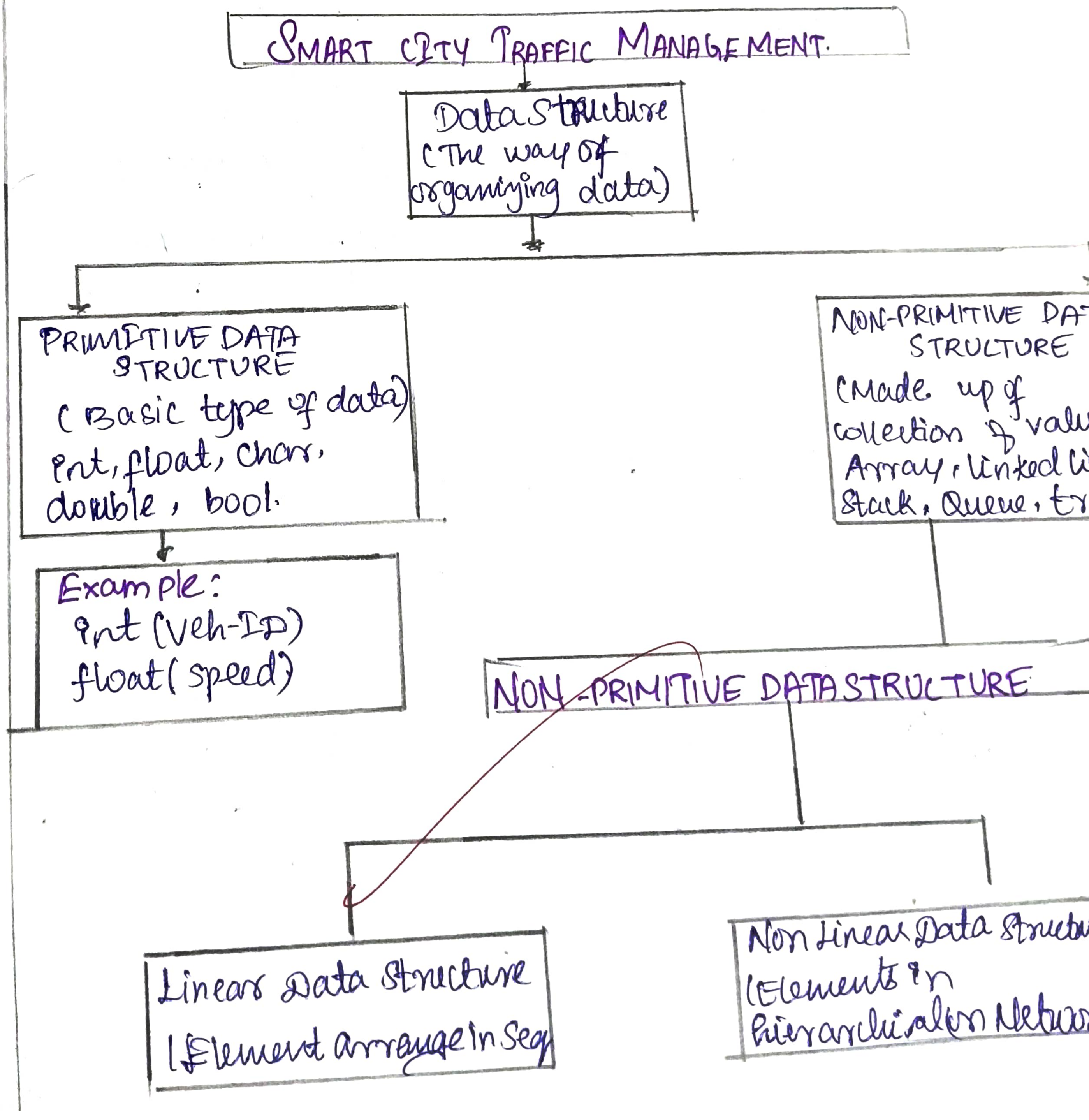
Operation
• Insert Pat
• Delete rec
• Search Pat
• Update pat

Time complex
• Stack : O(1)
• Queue : O(1)
• List Search : O(1)

Space complexity
• Dependent on no. of. Patient(n)

Performance
→ Fast retrived time
Efficient data handling

8] A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non Primitive Data Structures → Linear → Non Linear Structures. Includes examples such as array, linked lists, trees, and graphs and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

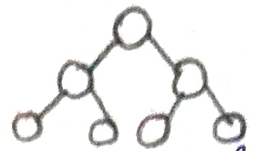


ARRAY

→ Fixed Size of similar element

eg: count[10];

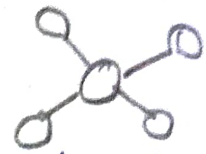
TREE



→ Hierarchial Structure.

eg: Road hierarchy

GRAPH



→ Network of node connect by edges.

Linked List

→ Dynamic collection of elements

eg: Store list of vehicles detect by Sensors.

Queue.

(FIFO)

Used for waiting lines.

Eg: vehicles wait at traffic signals

APPLICATIONS IN SMART CITY TRAFFIC.

VEHICLE MONITORING

• Store and process vehicle data from sensor

TRAFFIC SIGNAL CONTROL

• Queue data helps signal scheduling

ROUTE OPTIMIZATION

Graph help in find shortest / fastest.

CONGESTION DETECTION.

Analyze traffic density using array & graphs.